Engineering Mathematics - I · Unit_04 · Unit 4

# DRAFT — UE26MA141A · Unit_04 · Unit 4 · Definition of Beta function and its properties

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents

1. 1 · Definition of Beta function and its properties
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

# Definition of Beta function and its properties

## 1 Definition of Beta function and its properties

Study focus: Definition of Beta function and its properties. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000014 · EV0000015

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

In last class you have studied definition, properties and graph of gamma function.

**Key details**

- In this class, we are going to study definition of beta function and properties of beta function.
- Beta function is a definite integral whose integrant depends on two variables.
- It is also known as Euler's integral of first time.
- Beta function is defined as beta of m comma n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 is, where m and n both are positive.
- Let me call this as equation number one.
- We can also have an alternate form for beta function.
- We can get it by putting x is equal to sine square theta in equation number 1.
- When x is equal to sine square theta, dx is equal to two sine theta cos theta d theta.
- Also, the limits of integration change with the substitution.
- When x is equal to 0, that implies sine square theta is equal to 0, that implies theta is equal to 0.
- When x is equal to 1, that means sine square theta is equal to 1, or theta is equal to pi by 2.
- With this, we will get beta of m comma n as integral 0 to pi by 2, sin square theta all power m minus 1.
- One minus sine square theta whole power n minus one into two sine theta cos theta b theta.
- 2 is a constant we can keep outside the integral sign.
- Sin square theta wall power m minus 1 is sin power 2 m minus 2 theta.
- One minus sin square theta we know it is cos square theta.

- Beta of m comma n is now two times integral 0 to pi by 2 sin power 2 m minus 1 theta cos power 2n minus 1 theta d theta.
- This is another form of beta function, we can also call it as trigonometric form of beta function.
- Both the forms are important, so you can remember both the forms here.
- Both are useful when solving the problems.
- We can choose any one of these forms according to our convenience.
- Next, we will study the properties of beta function.
- First property is symmetry, which states beta of m comma n is equal to beta of L comma n.
- We can prove this by using the definition of beta function, which states beta of m, n is equal to integral 0 to 1 x power m minus 1 into 1 minus x power n minus 1 dx.
- then dx becomes minus of dt.
- Also the limits of integration change with the substitution x equal to 1 minus t implies t is equal to 1 minus x.
- Therefore, when x is equal to 0, t is equal to 1.
- And when x is equal to 1, right, t is equal to 0.
- Therefore, now we can have beta of m comma n is equal to integral 1 to 0 1 minus t power m minus 1 t power n minus 1 minus t.
- T power n minus 1 minus t to the power m minus 2 t, which is nothing but equal to beta of n comma m.
- Thus we have proved now beta of m comma n is exactly same as beta of n comma m.
- And now we know beta of m n is same as beta of n m, so we can also write beta of m, comma n is equal to 2 times integral 0 to pi by 2 cos power 2 m minus 1 theta into sine power 2n minus 1 theta d.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

To prove this, consider the diagnometric form of beta function that is beta of m, comma n is equal to 2 times integral 0 to pi by 2 sin power 2m minus 1 theta cos power 2n minus 1 theta d theta.

**Key details**

- Take 2m minus 1 is equal to p and 2n minus 1 is equal to q.
- Then we get m is equal to p plus 1 by 2 and n is equal to q plus 1 divided by 2.
- According to the definition of beta function, we should have m in beta of m comma n, m and n both are positive.
- So, m is greater than 0 implies now p plus 1 divided by 2 is greater than 0, that means p is greater than minus 1.
- This is what the conditions mentioned in property number 2.
- Therefore, integral 0 to pi by 2 sine power p theta cos power q theta d theta is equal to 1 by 2 times beta of p plus 1 by 2, comma q plus 1 divided by 2.
- This is also an important property.

- Whenever we come across the integrals in the form integral 0 to pi by 2, sin power p theta, cos power q theta, always we can express them in terms of beta functions.
- Already we proved now that integral 0 to pi by 2 sin power p theta cos power q theta d theta is 1 by 2 times beta of p plus 1 by 2 comma q plus 1 by 2.
- Then we get integral 0 to pi by 2 sin power p theta is e theta is equal to 1 by 2 times beta of p plus 1 by 2 comma 0 plus 1 divided by 2.
- Which is equal to 1 by 2 times beta of P plus 1 by 2, comma 1 by 2.
- And also we know that beta of m comma n is equal to beta of n, comma m.
- Thus we can conclude integral 0 to pi by 2 sine power p theta d theta is same as integral 0 to pi by 2 cos power p theta d theta and that is equal to half into beta of p plus 1 by 2 comma 1 by 2.
- Next property says that we can also express beta function in the form of an improper integral 0 to infinity x power m minus 1 divided by 1 plus x power m plus n dx.
- This property is also very useful because whenever we want to evaluate the integrals in the form integral 0 to infinity x power m minus 1 by 1 plus x power m plus n dx, always we can write them as beta of m comma n and we can find the value of beta of m comma n easily.
- Let me call this as equation number 1.

**Example or application from the lecture**

- In this let us put q is equal to 0.
- To prove this property, first let us consider the definition of beta function which says beta of m, n is equal to integral 0 to 1 x power m minus 1 minus x to the power n minus 1 bx.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

That means dx is equal to minus 1 divided by 1 plus y the whole square dy.

**Key details**

- That implies y is equal to infinity.
- When x is equal to 1, 1 plus y is equal to 1, that means y is equal to 0.
- into 1 minus 1 by 1 plus y whole to the power n minus 1 into minus dy divided by 1 plus y the whole square, which is equal to minus of integral infinity to 0.
- 1 divided by 1 plus y whole power m minus 1 into 1 plus y minus 1 divided by 1 plus y whole to the power n minus 1.
- which is equal to integral 0 to infinity y power n minus 1 divided by 1 plus y to the power m minus 1 into 1 plus y to the power n minus 1 into 1 plus y whole square into dy.
- That is m plus n divide.
- Because since we are dealing with the definite integrals.
- X and Y both are dummy variables.
- We can use any variable here.
- Remember this, this is also an important result.
- This can be proved very easily by considering the definition of beta function.

- So first term becomes now integral 0 to 1 x power m minus 1 minus x to the power n plus 1 minus 1 dx.
- And second terms after applying bit of m per m formula, so we will get integral 0 to 1 x power m plus 1 minus 1 into 1 plus x to the power n minus 1 dx.
- That is equal to integral 0 to 1 x power m minus 1 minus x to the power n dx plus integral 0 to 1 x power m into 1 minus x to the power n minus 1 dx.
- 1 minus x power n minus 1 into x power m minus 1 is the common factor.

**Example or application from the lecture**

- Next consider the fourth property that says beta of m comma n can be written as beta of m comma n plus one plus beta of m plus one comma n.
- Let us use this definition.
- First let us consider the right hand side.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

**Core explanation · 15:00–19:59**

Take it outside.

**Key details**

- That is nothing but equal to beta of m comma n.
- Thus we have beta of m comma n is equal to beta of m comma n plus 1 plus beta of m plus 1 comma n.
- With this we will end this session.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Definition of Beta function and its properties transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Definition of Beta function and its properties transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Definition of Beta function and its properties transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps. |
| Definition of Beta function and its properties transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/002_Definition_of_Beta_function_and_its_properties__c1ec05b61754baa1.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

### Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00006 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare... |
| SRC00007 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00008 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare... |
| SRC00009 | 4 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare... |

# C Revision and retrieval practice